



Digital twin-based prediction of vortex-induced vibration of a twin-box bridge deck within the lock-in region

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ABSTRACT

Wind tunnel tests and computational fluid dynamics (CFD) simulations are the two most common methods for investigating vortex-induced vibration (VIV) of a bridge deck at its design stage. However, current practice and research indicate that the two methods have their own uncertainties and drawbacks. This study thus integrates the two methods to develop global digital twins for investigating the VIV of a bridge deck section within its lock-in region. A twin-box bridge deck is first tested in a wind tunnel as a physical entity, and the relevant VIV information is collected. A virtual model for the physical entity is established using CFD simulation. The virtual model is then mapped with the physical entity through fusing the measurement data from the physical model with an optimization process, thereby establishing a local digital twin of the deck section for a given wind speed. Global digital twins are subsequently created based on local digital twins and the Kriging interpolation. The local and global digital twins are employed to investigate and eliminate blockage effects, and to predict the VIV of the deck section at different damping levels and wind speeds within the lock-in region. The results obtained demonstrate that the developed digital twins can provide more information and more accurate prediction of VIV of the deck section, compared with that provided by wind tunnel tests or CFD simulations alone.

Nomenclature

$a(t), v(t), y(t)$	Acceleration, velocity, and displacement at time t , m/s^2 , m/s , m
ADi	Additional damping case i
B	Width of the deck section, mm
$c(y)$	Mechanical damping coefficient at vibration amplitude y , $N \cdot s/m$
$c_0(y)$	Nominal damping coefficient at vibration amplitude y , $N \cdot s/m$
CFD	Computational fluid dynamics
$f_0(y)$	Nominal vibration frequency at vibration amplitude y , Hz
$f_{0,NAD}(y)$, $f_{0,ADi}(y)$	Nominal vibration frequency in NAD or ADi case at vibration amplitude y , Hz
f_0^s, f_0^m	Simulated and measured VIR frequency, Hz
$F(t)$	Aerodynamic lift force, N
$k_0(y)$	Stiffness coefficient of the deck section at vibration amplitude y , N/m
m	Mass, kg
m_0	Air-induced mass, kg
NAD	Damping system without additional damping level
R_r	Relative error
VIF	Vortex-induced force, N
VIR	Vortex-induced response
VIV	Vortex-induced vibration

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y, y^s, y^m	Vibration amplitude, simulated and measured VIR amplitude, mm
$\xi(y)$	Mechanical damping ratio at vibration amplitude y
$\xi_{air}(y)$	Air-induced damping ratio at vibration amplitude y
$\xi_0(y)$	Nominal damping ratio at vibration amplitude y
$\xi_{0,NAD}(y)$, $\xi_{0,ADi}(y)$	Nominal damping ratio in NAD or ADi case at vibration amplitude y

1. Introduction

Wind moving across a bridge deck will generate periodic vortex shedding. When the shedding frequency matches one of the natural frequencies of the bridge deck, vortex-induced vibration (VIV) occurs. VIV of a bridge deck exhibits distinct characteristics such as lock-in region, aeroelastic interaction, and amplitude-limited vibration (Scruton, 1963; Simiu and Scanlan, 1978). Wind tunnel tests and computational fluid dynamics (CFD) simulations are the two most common methods nowadays to investigate and mitigate, if necessary, VIV of the bridge deck at its design stage. Nevertheless, significant VIV still occurred in

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some of the in-service long-span bridges, indicating that there are some uncertainties and drawbacks in the currently used two methods.

Wind tunnel tests are intuitive and effective methods for exploring VIV characteristics and responses of bridge deck sections (Yang et al., 2016; Yuan et al., 2017; Zhu et al., 2017a). However, there are some uncertainties and constraints involved in the wind tunnel tests (Klamo et al., 2006; Gao and Zhu, 2015; Zhang and Xu, 2019). For instance, the mechanical damping and stiffness coefficients of a spring-suspended bridge deck section in a wind tunnel are difficult to be accurately identified because of their possible nonlinear characteristics and the air-induced aerodynamic forces acting on the deck section (Cao and Ge, 2017). The blockage effects also exist in wind tunnel tests, which may affect the free development of VIV (Morse et al., 2008; Li et al., 2018). Moreover, it is difficult to visualize flow field around a deck section and measure the aeroelastic forces directly on the deck section during VIV.

The CFD simulation of VIV of a deck section could overcome some of the difficulties encountered by wind tunnel tests (Martini et al., 2021; Camera et al., 2023). However, the accuracy of CFD simulation results closely relies on the selection of turbulence model, the quality of meshing scheme, and the time-step of iteration. The complex auxiliary facilities installed on the deck section are often difficult to be fully simulated in CFD. It seems that researchers rarely realize the importance of establishing a mirror numerical model for the physical model and always simplify the physical characteristics of and the flow field around the deck section during the construction of numerical model (Sarwar and Ishihara, 2010; Kavrakov and Morgenthal, 2018; Noguchi et al., 2020). These deficiencies lead to the inconsistency in the VIV characteristics and responses between CFD simulation and wind tunnel test (Dong et al., 2021; Zhang et al., 2023; Zhao et al., 2024). CFD simulations and wind tunnel tests are not inherently integrated at present in investigating VIV of a deck section.

The recently developed digital twin technology provides a novel and effective method to enhance the prediction accuracy and reliability in bridge engineering by offering a synergistic combination between the physical model and the virtual model. Lin et al. (2021) developed a digital twin for collapse fragility assessment of cable-stayed bridges subjected to strong earthquakes. Lao et al. (2024) established a digital twin for the prediction of microcrack initiation and propagation in the weld of orthotropic steel deck. Ye et al. (2024) used the digital twin technology to identify the crystal plasticity material parameters by integrating the multiscale finite element model with the physical tests. In the field of bridge wind engineering, the digital twin technology was also used to provide the prediction of VIV of cable-supported bridges at their operation stage (Zhang et al., 2024) and at their design stage (Li et al., 2024a). Nevertheless, although the feasibility of the digital twin established for VIV of a streamlined bridge deck section was demonstrated by Li et al. (2024a), there are some deficiencies in their study: (1) the uncertainties in the wind tunnel tests of VIV are not clearly clarified; (2) the particle swarm optimization (PSO) algorithm used to develop digital twins consumes large computational resources; and (3) the digital twin was established at a particular wind speed, which did not cover the lock-in region of VIV.

To overcome these deficiencies, this study develops global digital twins for the VIV of a deck section within the lock-in region. A twin-box deck section suspended by springs and tested in a wind tunnel is taken as a physical model while a numerical model established in ANSYS FLUENT is taken as a virtual model. A series of local digital twins are first developed by mingling the virtual model with the physical model using the multivariate polynomial regression algorithm. Global digital twins are then established using the local digital twins and the Kriging interpolation. The remaining parts of this paper are as follows. Section 2 describes the framework for digital twin-based prediction of VIV of a bridge deck. Section 3 introduces the wind tunnel tests of the physical deck section, the data collection from the physical deck section, and the uncertainties involved in the physical deck section during VIV test. Section 4 provides the details of establishing a virtual model and

confirms the necessity of establishing a digital twin for VIV of a twin-box deck section. Section 5 introduces the detailed procedure of developing a local digital twin for VIV under a particular wind speed using the multivariate polynomial regression algorithm. Section 6 utilizes these established local digital twins to explore the blockage effect on VIV and predict the VIR amplitudes and frequencies of the deck section. Section 7 uses the local digital twins and the Kriging interpolation to develop global digital twins and uses the global digital twins to predict the VIR of the deck section of other mechanical damping levels within the lock-in region. Section 8 provides a summary of the conclusions from this study.

2. Framework for digital twin-based prediction of VIV of a bridge deck

The digital twin has the four essential elements in order (Grieves and Vickers, 2017): (a) physical entity and data collection from the physical entity; (b) simulation of a physical entity to have a virtual entity; (c) interaction and mapping between the physical and virtual entities to form a digital twin; and (d) application of and prediction by the digital twin. Accordingly, the procedure for establishing a framework for digital twin-based prediction of vortex-induced vibration of a twin-box bridge deck within the lock-in region in this study is summarized as follows and shown in Fig. 1:

- (1) A twin-box bridge deck and its wind tunnel test are selected as the physical entity. Vortex-induced vibration (VIV) of the bridge deck is investigated in a wind tunnel. The information on the flow field characteristics of the wind tunnel and the dynamic characteristics, VIV responses, and vortex-induced forces (VIF) of the bridge deck are collected. These data are further analyzed to identify uncertainties in the wind tunnel tests. These contents are presented in Section 3.
- (2) A virtual model of the physical entity is established in ANSYS FLUENT. The model is consistent with the physical entity and the test conditions in terms of aerodynamic shape, computational domain, boundary conditions, and flow characteristics. Furthermore, VIV information collected from the wind tunnel tests is fused into the model via UDFs, enabling deeper data integration. The virtual model should also adopt an appropriate turbulence model and ensure grid and time-step independence. This part is detailed in Section 4.
- (3) For the VIV of the bridge deck at a given wind speed, the virtual model is iteratively updated through data fusion and optimization algorithm to eliminate uncertainties in the physical entity and update the virtual model, thereby establishing a local digital twin. During this process, optimization parameters, objective function, and optimization algorithm are defined, and the vibration responses measured by sensors in the wind tunnel are continuously used to correct the numerical simulation results, ensuring consistency with the test data. By this method, local digital twins are established within the lock-in region. The procedure and implementation are described in Section 5.
- (4) The local digital twins can be directly applied in two aspects: estimating blockage effects in wind tunnel tests to provide more accurate VIV predictions; and identifying air-induced added mass and damping as well as mechanical damping and stiffness of a bridge deck in the wind tunnel tests to enable precise prediction of VIV responses under different damping ratios. These applications are detailed in Section 6.
- (5) Based on the local digital twins, a global digital twin is developed using Kriging interpolation to enable rapid prediction of VIV amplitudes under any wind speed within lock-in region. By further incorporating local digital twins from different damping systems as samples, the global digital twin is extended to account for damping coefficients, stiffness coefficients, and wind speeds. The methodology and applications are presented in Section 7.

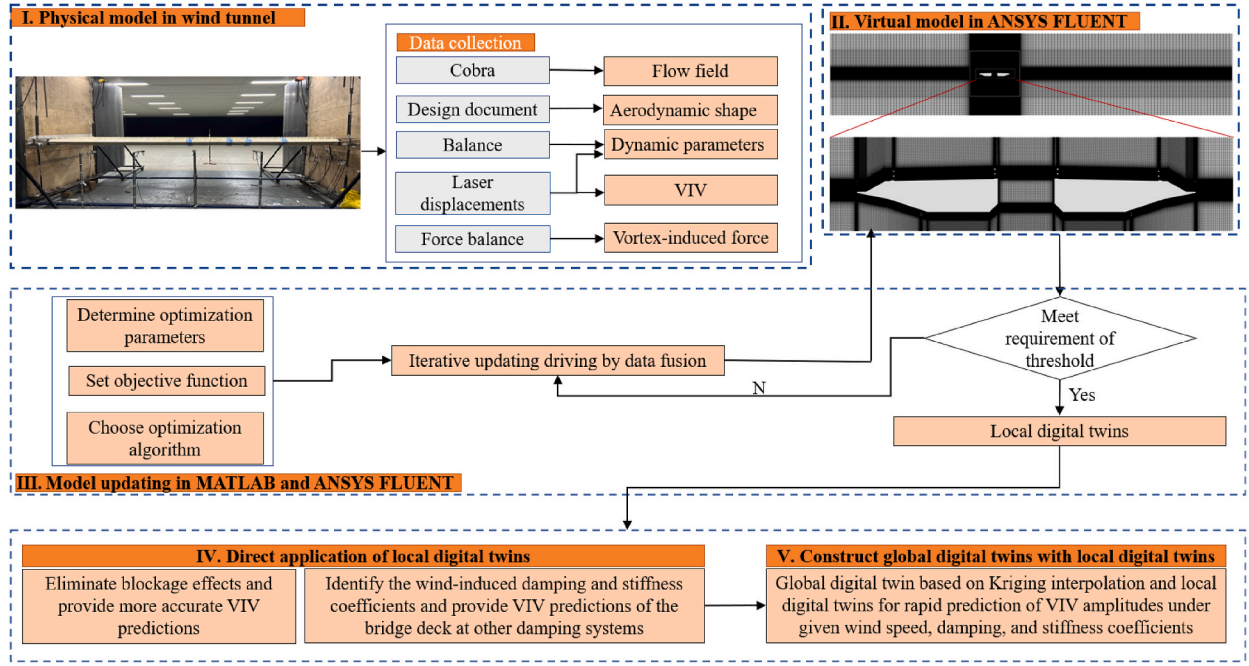


Fig. 1. Flowchart of establishing a global digital twin for the VIV of the bridge deck and its application.

3. Physical model and uncertainties in wind tunnel tests

The bridge deck in wind tunnel tests serves as the physical model in the digital twin framework. It provides the test data of VIV, including the VIV responses and VIF, which are required for the subsequent mapping with the virtual model.

3.1. Physical model and measurement system

The physical model used in this study is a deck sectional model of the Xihoumen Bridge tested in the XNJD-3 wind tunnel at Southwest Jiaotong University. This wind tunnel is a closed-circuit wind tunnel,

measuring 36m (length) by 22.5m (width) by 4.5m (height) and with a maximum mean wind speed of 16.5 m/s. The longitudinal and lateral turbulence intensities of the smooth flow within the wind tunnel were measured at 0.68 % and 0.72 %, respectively.

The bridge deck section was scaled at a geometric ratio of 1:20: the deck section had a width of 1700 mm, a height of 175 mm, a length of 3900 mm, and a central slot width of 300 mm, as shown in Fig. 2a. The deck section comprised an upstream box girder, a downstream box girder, and four crossbeams connecting the two box girders (see Fig. 2b). The deck section was suspended by eight springs from the fairing walls at both ends of the deck and the fairing walls were used to support the VIV of the deck section in the vertical direction and maintain a two-

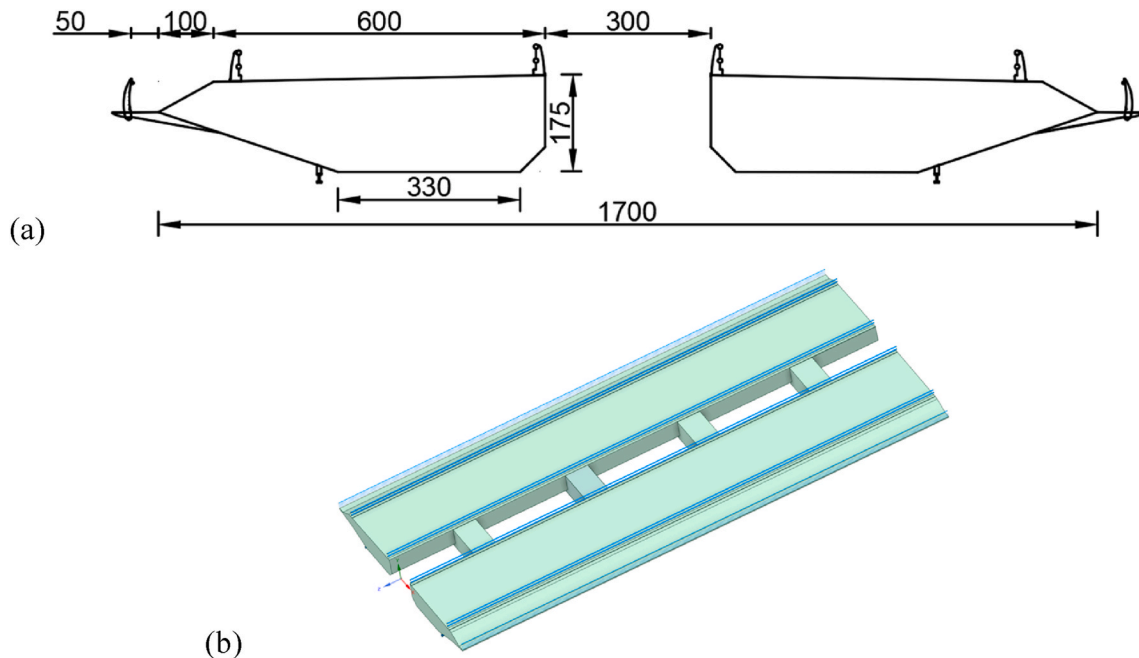


Fig. 2. Bridge deck: (a) sectional dimension (unit: mm), (b) schematic diagram (Li et al., 2024b).

dimensional flow field as close as possible, as shown in Fig. 3a.

For measurements, a Cobra probe positioned 1.5 m from the deck section at the same elevation recorded incoming wind speeds (see Fig. 3a). The vortex-induced response (VIR) time histories of the bridge deck were recorded by five laser displacement transducers and one accelerometer in the vertical direction because only vertical VIV occurred. Eight force balances with four pairs of force-measuring strips were used to measure the aerodynamic forces acting on the strips (see Fig. 3b). The measured aerodynamic force consisted of vortex-induced force (VIF) and air-induced aerodynamic forces. The sampling frequency of all the sensors and force balances is 1024 Hz. More detail information on the wind tunnel tests and the measurement system can be found in the reference (Li et al., 2024b).

3.2. Dynamic characteristics of physical model and uncertainties

The mechanical parameters such as the mass, damping coefficient, and stiffness coefficient of the deck section are the most important factors for the dynamic characteristics and VIV of the deck section after its aerodynamic shape is determined. The deck section mass can be directly weighted through the electronic scales as 200.2 kg or 51.3 kg/m. For the spring-suspended deck section installed in the wind tunnel for VIV, the dynamic characteristics (mechanical damping ratio and vibration frequency) are often assumed to be constant (Zhu et al., 2017b; Wu and Laima, 2021; Gao et al., 2022). Based on this assumption, the damping ratio and vibration frequency of the deck section are then identified from the free-decay time history obtained in the wind tunnel without incoming wind, as shown in Fig. 4. The damping coefficient (c_0) and stiffness coefficient (k_0) of the deck section are finally determined using the following equations:

$$c_0 = 4\pi f_0 \xi_0 m \quad (1a)$$

$$k_0 = 4\pi^2 f_0^2 m \quad (1b)$$

where f_0 is the vibration frequency identified from the free-decay time history without incoming wind; ξ_0 is the damping ratio identified from the free-decay time history without incoming wind; and m is the mass of the spring-suspended bridge deck.

However, it was found out that the damping ratio and vibration frequency identified from the free-decay time history depend on vibration amplitude and therefore they are not constant (Zhu et al., 2013; Cao and Ge, 2017; Gao and Zhu, 2017; Noguchi et al., 2020; Duan et al., 2023). It was then suggested that the damping and stiffness coefficients of the deck section be determined by Eq. (2):

$$c_0(y) = 4\pi f_0(y) \xi_0(y) m \quad (2a)$$

$$k_0(y) = 4\pi^2 (f_0(y))^2 m \quad (2b)$$

where y is the vibration amplitude in the free-decay time history.

It was suggested that for VIV of the deck section, the vibration amplitude y should be selected roughly equal to the maximum VIV amplitude. Clearly, the nonlinearities in the damping ratio and vibration frequency cause uncertainties in the calculated damping and stiffness coefficients of the deck section. Furthermore, it was pointed out that the damping ratio identified from the free-decay time history includes two components: the mechanical damping ratio and the air-induced damping ratio (Cao and Ge, 2017; Zhang and Xu, 2019). Thus, the damping ratio $\xi_0(y)$ and the corresponding damping coefficient $c_0(y)$ in Eq. (2a) are called the nominal damping ratio and the nominal damping coefficient (Gao and Zhu, 2015). To make a distinction, the mechanical damping ratio and the corresponding mechanical damping coefficient are denoted as $\xi(y)$ and $c(y)$, respectively. Air-induced damping is one kind of aerodynamic damping introduced by the interaction between the vibrating bridge deck and its surrounding air. The characteristics of air-induced damping are complex. It was found that the air-induced damping is related to the aerodynamic shape and vibration amplitude of the bridge deck, and it takes a dominant proportion of the nominal damping compared with the mechanical damping in some cases (Cao and Ge, 2017). Therefore, if the air-induced damping ratio and damping coefficient are denoted as $\xi_{air}(y)$ and $c_{air}(y)$, respectively, the following relationships exist:

$$\xi_0(y) = \xi(y) + \xi_{air}(y) \quad (3a)$$

$$c_0(y) = c(y) + c_{air}(y) \quad (3b)$$

Due to the difficulty in accurately measuring the air-induced damping ratio, the mechanical damping ratio $\xi(x)$ is hardly to be extracted from the identified nominal damping ratio. Moreover, in addition to the air-induced damping, the air-induced mass m_0 is also introduced during VIV of a bridge deck in a wind tunnel. The air-induced mass will affect the vibration frequency of the deck. Therefore, the relationship between the nominal damping coefficient and the nominal damping ratio, and the relationship between the nominal vibration frequency and the nominal stiffness coefficient should be expressed by Eq. (4) other than Eq. (2):

$$c_0(y) = 4\pi f_0(y) \xi_0(y) (m + m_0) \quad (4a)$$

$$k_0(y) = 4\pi^2 (f_0(y))^2 (m + m_0) \quad (4b)$$

Again, the air-induced mass of the deck section is difficult to measure. As a result, the nominal damping and stiffness coefficients of the spring-suspended bridge deck are difficult to be determined although

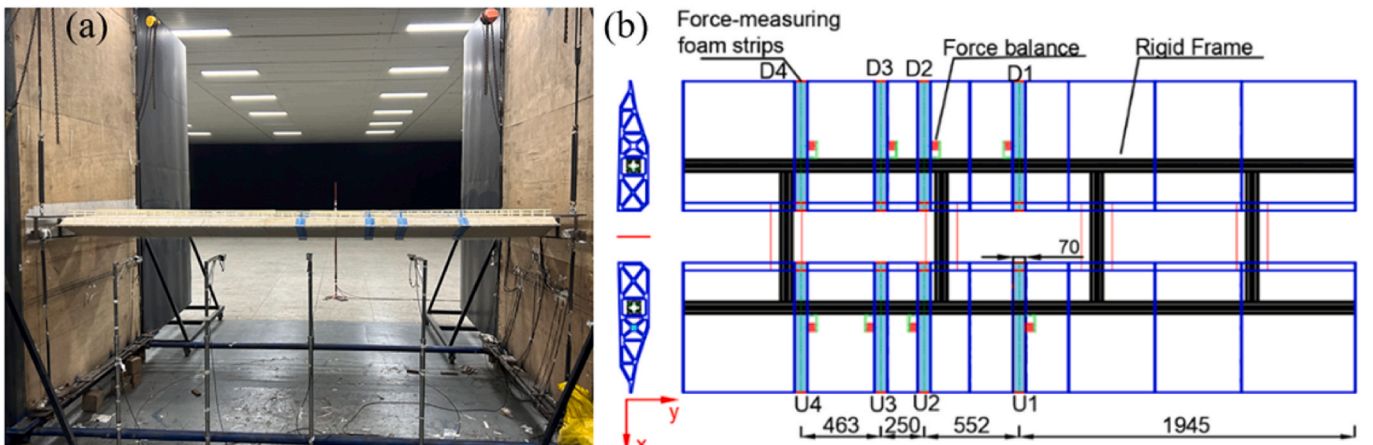


Fig. 3. Installation and configuration of deck section: (a) installation, (b) configuration (unit: mm) (Li et al., 2024b).

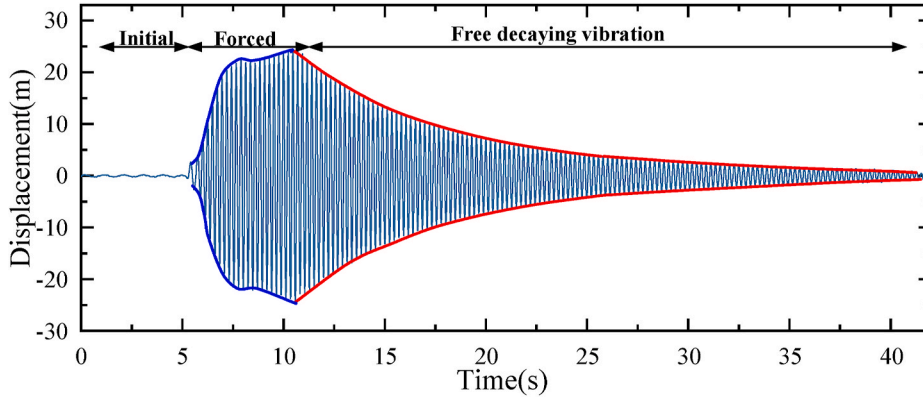


Fig. 4. Free-decay time history of deck section within a wind tunnel without incoming wind.

the nominal damping ratio and vibration frequency are identified. Furthermore, because of the dependence of the nominal damping ratio and nominal vibration frequency on vibration amplitude as well as the existence of air-induced mass and damping, the mechanical damping and stiffness coefficients cannot be accurately identified from wind tunnel tests. The effects of the air-induced mass and damping on VIV of the deck section cannot be assessed appropriately. It is noted that the above discussions refer to the most used wind tunnel test technique as described above, which excludes some special wind tunnel test techniques (Cao and Ge, 2017; Li et al., 2021) that can be used to determine the mechanical damping and stiffness coefficients of the deck section.

In this study, the bridge deck suspended by several springs was tested in the wind tunnel without incoming wind to obtain the free-decay time history and identify the nominal damping ratios and vibration frequencies at different initial displacements. There were four mechanical damping levels realized by changing the number of elastic ropes wrapped around the springs. The four mechanical damping levels were labeled NAD (no additional elastic ropes/no additional damping), AD1 (with additional damping case 1), AD2 (with additional damping case 2), and AD3 (with additional damping case 3). Fig. 5 shows the variations of the nominal damping ratio and vibration frequency of the deck section with vibration amplitude (2.5 mm above) for the four mechanical damping levels. It is seen that for each mechanical damping level, the nominal damping ratio increases considerably and linearly with increasing amplitude, while the nominal vibration frequency decreases slightly and nonlinearly with vibration amplitude. Moreover, referring to the previous wind tunnel tests on VIV of twin-box decks (Laima and Li, 2015; Liu et al., 2021; Chen et al., 2022), the damping ratios used in this study cover a wider range than the previous ones.

By fitting the test results, the linear relationship between the nominal damping ratio and vibration amplitude can be expressed by Eq. (5a) to Eq. (5d) for the mechanical damping levels NAD, AD1, AD2, and AD3, respectively. It is seen that the constant value in the equations increases from Eq. 5 (a) to Eq. (5d) but the coefficient of y remains almost the same when the vibration amplitude of the deck section exceeds 2.5 mm.

$$\xi_{0,NAD}(y) = 0.00344 + 0.0001116y \quad (5a)$$

$$\xi_{0,AD1}(y) = 0.00557 + 0.0001144y \quad (5b)$$

$$\xi_{0,AD2}(y) = 0.00648 + 0.0001163y \quad (5c)$$

$$\xi_{0,AD3}(y) = 0.00762 + 0.0001174y \quad (5d)$$

Eq. 6 (a) to Eq. (6d) are the fitted equations for the nominal vibration frequency and vibration amplitude for the deck section with the mechanical damping levels NAD, AD(1), AD(2), and AD(3), respectively. It is seen that the constant value and the coefficient of y^2 decrease slightly from Eq. 6 (a) to Eq. 6 (d) and the coefficient of y has a little difference.

$$f_{0,NAD}(y) = 4.314 - 0.00335y + 0.0000794y^2 \quad (6a)$$

$$f_{0,AD1}(y) = 4.297 - 0.00318y + 0.0000772y^2 \quad (6b)$$

$$f_{0,AD2}(y) = 4.287 - 0.00321y + 0.0000735y^2 \quad (6c)$$

$$f_{0,AD3}(y) = 4.275 - 0.00311y + 0.0000704y^2 \quad (6d)$$

Nevertheless, there are uncertainties in using the above nominal damping ratios and nominal vibration frequencies to assess the accuracy

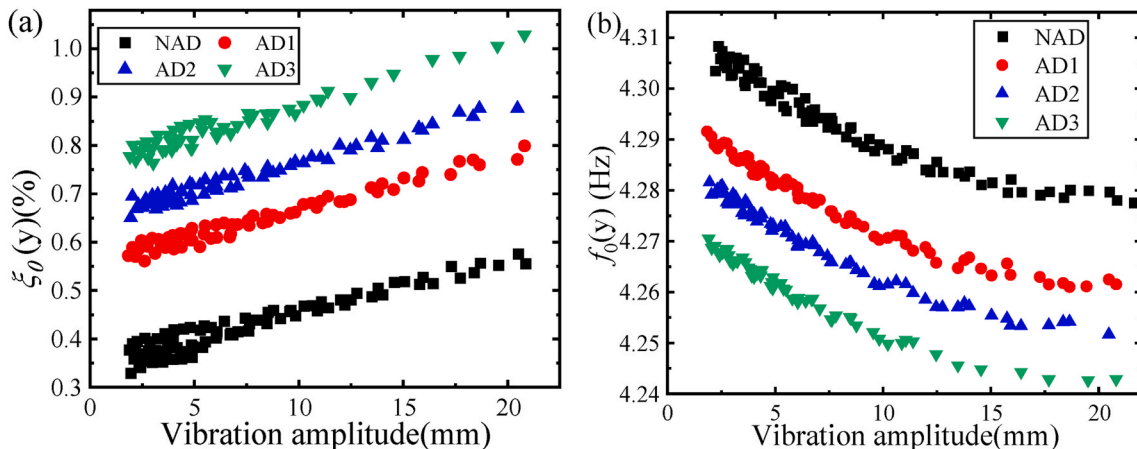


Fig. 5. Variations with vibration amplitude: (a) nominal damping ratios, (b) nominal vibration frequencies.

of wind tunnel test results of VIV of the deck section and convert the wind tunnel test results to the prototype bridge. The digital twin technology is thus sought in this study to solve this problem to some extent.

3.3. Test data collection

The relevant test data of the bridge deck should be collected for constructing its digital twin. The vertical VIR amplitudes of the bridge deck under several wind speeds, obtained from the wind tunnel tests, are shown in Fig. 6 for the four mechanical damping levels. It is seen that for each damping level, there is a lock-in region, in which the vortex-induced displacement amplitude increases from zero to its maximum and then decreases to zero as wind speed increases. It is also seen that higher mechanical damping level leads to a lower displacement amplitude, a narrower lock-in region, and a higher wind speed at which the maximum amplitude occurs. Specifically, the measured maximum displacement amplitude is 12.89 mm at 5.14 m/s for NAD, 9.65 mm at 5.28 m/s for AD1, 7.94 mm at 5.35 m/s for AD2, and 6.57 mm at 5.42 m/s for AD3. The corresponding total damping ratios of 0.50 %, 0.66 %, 0.73 %, and 0.84 % further demonstrate the nonlinear relationship between vibration amplitude and damping ratio.

For each damping level and each wind speed within the lock-in region tested in the wind tunnel, the VIR and VIF time histories of the deck section were simultaneously recorded. Fig. 7 shows the recorded VIR and VIF time histories under a wind speed of 5.28 m/s for AD1. It is seen that the steady VIR time history looks like a harmonic vibration, while the VIF time history seems a harmonic excitation. In this study, the VIR time histories will be used for the establishment of digital twins, whereas the VIF time histories will be used to validate the established digital twin.

The VIF time histories are also used to find out a wind speed range within the lock-in region, which is suitable for developing a 2D virtual model using CFD simulation. There were four sets of force-measurement strips positioned in the physical model at six spacing intervals: 0.25 m, 0.463 m, 0.551 m, 0.713 m, 0.801 m, and 1.264 m, as shown in Fig. 3b. The spanwise correlation ratios ρ_{ij} of VIFs at locations i and j are calculated for the deck section with NDA under different wind speeds, and the results are shown in Fig. 8. It is seen that the coherence of VIFs across different strips ranges from 95 % to 99.8 %, except for the two specific wind speeds at which VIV either starts or ends. Therefore, the 2D virtual model can be used for investigating VIV at all the wind speeds within the lock-in region, except for the two specific wind speeds where VIV just starts or ends.

In the subsequent sections, the test data collected from VIV with NAD will be used to develop a virtual model and local digital twins. The deck section was tested under a total of 16 effective wind speeds within the lock-in region, as listed in Table 1 and denoted from No.1 to No.16. The

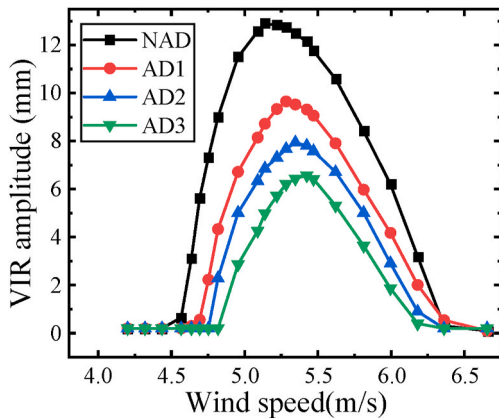


Fig. 6. Measured VIR displacement amplitudes and lock-in regions for different damping levels.

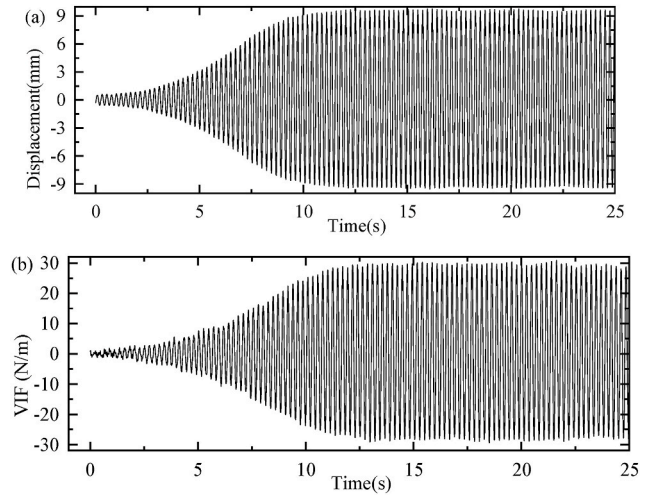


Fig. 7. The development of VIV under a wind speed of 5.28 m/s for AD1: (a) VIR, (b) VIF.

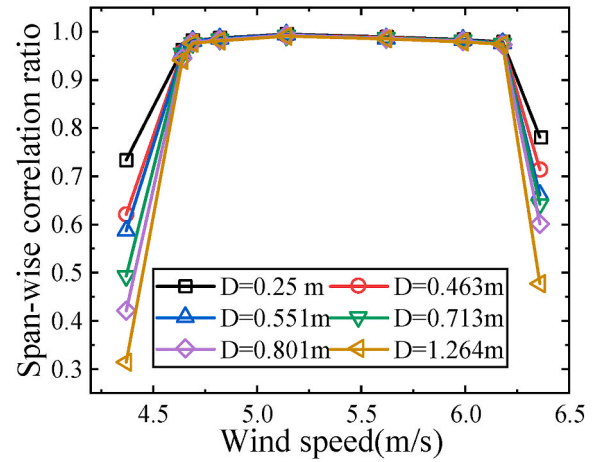


Fig. 8. Spanwise coherence of VIFs on deck section with NAD.

vertical VIR amplitude and VIR frequency under each wind speed are also shown in Table 1. The local digital twins will be constructed for the deck section under wind speeds No.2, No.4, No.5, No.7, No.9, No.11, No.13, and No.15, while the test data obtained at other wind speeds will be used to verify the accuracy of the global digital twins.

4. Establishment of virtual model

The virtual model provides the essential basis for constructing the digital twin. This section introduces how to establish a virtual model for the test deck section with NAD using CFD simulation. Then, the CFD simulation results from the virtual model are compared with the wind tunnel test results to highlight the necessity of updating the virtual model into local digital twins.

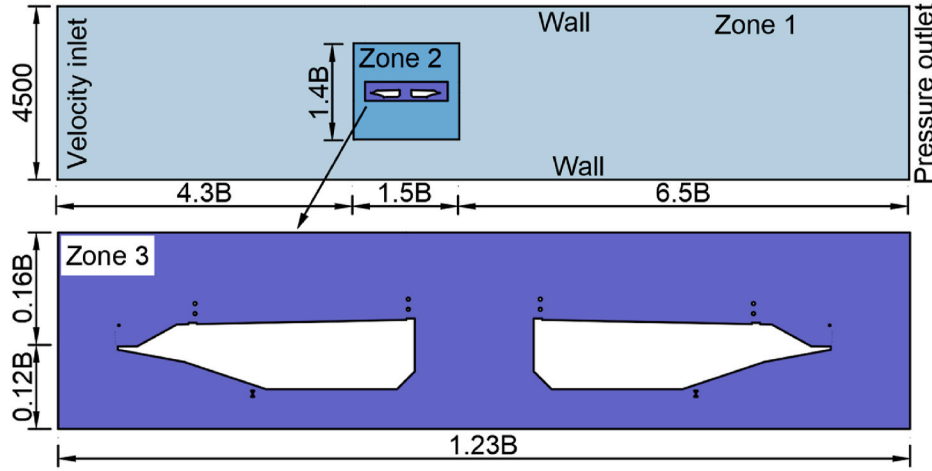
4.1. Virtual model

The size and dimension of the virtual deck section are the same as the physical deck section at a geometric scale of 1:20, as shown in Zone 3 of Fig. 9. The height and length of the computational domain are also the same as those of the wind tunnel with a height of 4.5 m and a length of 12.3B (where B is the width of the deck section). The computational domain shown in Fig. 9 includes three zones: Zone 1 is a stationary zone; Zone 2 is a deformative zone; and Zone 3 is a rigid zone. The virtual deck

Table 1

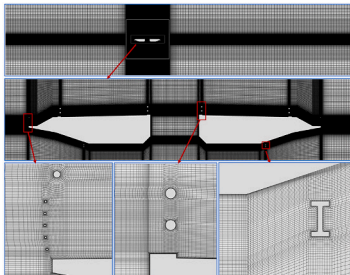
The test data of deck section with NAD under different wind speeds.

No.	Wind speed(m/s)	VIR amplitude (mm)	VIR frequency (Hz)	No.	Wind speed(m/s)	VIR amplitude (mm)	VIR frequency (Hz)
1	4.64	3.12	4.315	9	5.28	12.72	4.341
2	4.69	5.63	4.319	10	5.35	12.47	4.347
3	4.75	7.32	4.321	11	5.42	12.14	4.353
4	4.82	9.00	4.324	12	5.47	11.76	4.358
5	4.95	11.50	4.327	13	5.62	10.59	4.367
6	5.09	12.57	4.330	14	5.81	8.42	4.378
7	5.14	12.89	4.333	15	5.99	6.21	4.391
8	5.22	12.83	4.337	16	6.18	3.19	4.402

**Fig. 9.** Computational domain of virtual model (unit: mm).

section is set inside of the rigid zone to maintain the mesh quality around the deck section during VIV. The left side of the domain is defined as a velocity inlet, as shown in Fig. 9, with a wind speed tested in the wind tunnel. According to the flow field in the wind tunnel, the turbulent intensity and viscosity ratio are set to 0.7 % and 2, respectively. The right side of the domain is set as a pressure outlet with zero pressure. No-slip stationary wall is assigned to the upper and lower boundaries of the domain to simulate the boundary layer conditions of the wind tunnel.

The mesh sizes within the computational domain are crucial for ensuring simulation accuracy and efficiency. The thickness of the first layer meshes on the deck surface is set to 0.048 mm to ensure y^+ values below 1 at a wind speed of 6.5 m/s, where VIV disappears before wind speed reaches this level. The boundary layer meshes consist of 15 layers with a growth rate of 1.2 to keep the meshing quality. Additionally, the meshes around the rails and maintenance tracks are refined to improve simulation resolution. To optimize meshes numbers and improve computation efficiency, a variable mesh length strategy is implemented along the deck surface (see Fig. 10). In regions prone to flow separation and reattachment, such as wind fairings and corners, smaller mesh sizes are used along the incoming flow direction. In less critical areas, larger

**Fig. 10.** Mesh sizes of virtual model.

mesh sizes are used. This adaptive mesh sizing approach reduces the total grid count to 99249, significantly improving computation efficiency without losing accuracy. All used meshes are structured quadrilateral meshes, ensuring consistent quality and reliability in simulation results.

The flow field is resolved using the RANS-based SST $k-\omega$ turbulence model. The Simple C algorithm is employed to address pressure-velocity coupling within this turbulence model. A second-order upwind transient formulation is used for spatial discretization. The convergence of the governing equations is achieved by setting the residual to be less than 0.00001. The time interval is carefully set at 2×10^{-4} to keep the courant number below 1 during the simulation. The diffusion-based smoothing dynamic mesh technology, which consumes more computational resources than spring-based smoothing but provides larger deformations, is adopted for the movement of the deck section.

The weak coupling method is adopted to facilitate solving this fluid-structure interaction (FSI) problem. The procedure of weak coupling method for solving the vertical VIV of the deck section at each time-step can be concluded as follows:

- (1) Calculate and extract the aerodynamic lift force acting on the deck section by solving the RANS governing equation using the fluid solver.
- (2) Apply the aerodynamic lift force to the dynamic equation, Eq. (8), of the deck section and then solve the dynamic equation using the structural solver and the Newmark-beta method:

$$ma(t) + cv(t) + ky(t) = F(t) \quad (8)$$

where m , c , k represent the mass (unit: kg), mechanical damping coefficient (unit: N·s/m), and stiffness coefficient (unit: N/m) of the deck section, respectively; $a(t)$, $v(t)$ and $y(t)$ denote the vertical acceleration, velocity and displacement response at time t , respectively; $F(t)$ represents the aerodynamic lift force acting on the deck section (unit: N).

- (3) Update the position of the deck section in the CFD simulation based on the calculated velocity from the structural solver.

The fluid-structure interaction, that is VIV, of the deck section in the ANSYS FLUENT can be realized through repeating the above procedure during the simulation. It should be noted that the air-induced damping and air-induced mass are introduced by the interaction between the deck section and air so that they are included in $F(t)$ during the simulation. As a result, the c and k in Eq. (8) are the mechanical parameters, which are hardly identified from the wind tunnel test.

4.2. Independence of mesh size and time step

The VIV of the bridge deck involves complex phenomena like flow separation and reattachment, requiring finely tuned boundary layer meshes to accurately capture flow field details. The appropriateness of the boundary layer mesh size is typically assessed by the y^+ value, which quantifies the ratio of the distance from a grid node to the surface of structure relative to the characteristic scale of flow. A y^+ value less than 5 indicates that the flow resides within the viscous sublayer, where viscous forces dominate and velocity distribution is nearly linear. Under these conditions, the mesh is fine enough for the simulation requirements of the SST $k-\omega$ model. To improve computational accuracy, the mesh independence test is thus performed to ensure the quality of the CFD simulation. In this study, the mesh independence test is performed for the deck section at two critical wind speeds: 5.14 m/s, where the maximum VIR occurs, and 6.5 m/s, where VIV ceases (see Fig. 5). Four different mesh schemes are implemented with the first-layer boundary mesh height of 0.027 mm, 0.048 mm, 0.060 mm, and 0.072 mm. These correspond to the y^+ value of 0.75, 1, 1.25, and 1.5 at a wind speed of 6.5 m/s, respectively. Table 2 presents the simulation results, and the mesh scheme 3 is selected in considering both simulation accuracy and computation efficiency.

The three distinct time steps, 1×10^{-4} s, 2×10^{-4} s, and 4×10^{-4} s, are selected to evaluate the sensitivity of the simulated results to the time step. The simulated results with different time steps are listed in Table 3. The time step of 2×10^{-4} s is finally adopted in the subsequent simulations for both simulation accuracy and computation efficiency.

4.3. Comparison of CFD simulation with wind tunnel test

As indicated by Eq. (8), the mechanical damping and stiffness coefficients of the deck section should be known to perform the CFD simulation of VIV using the weak coupling method. However, as discussed in Section 2, it is difficult to obtain these two coefficients from the wind tunnel tests. Therefore, following the current practice and based on the nominal damping ratio, the nominal vibration frequency, and the VIR amplitude measured from the wind tunnel tests, the mechanical damping and stiffness coefficients are assumed and kept the constant values within the lock-in region for CFD simulations in this subsection.

Let us take the test deck section with NAD as an example and consider the two wind speeds: 4.64 m/s, No.1 wind speed listed in Table 1 with the minimum VIR, and 5.14 m/s, No.7 wind speed listed in Table 1 with the maximum VIR. Consider the vibration amplitudes of the deck section at the wind speeds of 4.64 m/s and 5.14 m/s are 3.12 mm

Table 2
Simulated results with different mesh schemes.

Mesh scheme	First layer height (mm)	Maximum y^+ (5.14/6.5 m/s)	VIR amplitude (mm) at 5.14 m/s	St at wind speed of 6.5 m/s	Mesh number
1	0.072	1.25/1.50	14.36	0.1742	93521
2	0.060	1.00/1.25	13.77	0.1714	96467
3	0.048	0.80/1.00	13.71	0.1703	99249
4	0.027	0.48/0.75	13.72	0.1704	104982

Table 3

Simulated results with different time steps.

Time step(s)	Maximum courant number for 5.14 m/s	Maximum courant number for 6.5 m/s	VIR amplitude (mm) at 5.14 m/s	St at wind speed of 6.5 m/s
4×10^{-4}	0.19	0.55	13.83	0.1709
2×10^{-4}	0.11	0.32	13.73	0.1703
1×10^{-4}	0.07	0.24	13.76	0.1704

and 12.89 mm, respectively. By using Eq.(5-a), the nominal damping ratios of the deck section at the wind speed of 4.64 m/s and 5.14 m/s are 0.379 % and 0.488 %, while the nominal vibration frequencies calculated by Eq.(6-a) are 4.325 Hz and 4.284Hz, respectively. Eq. (2) is then adopted to obtain the assumed mechanical damping coefficients of 41.235 N s/m and 52.592 N s/m and the assumed mechanical stiffness coefficients of 147831.8 N/m and 145042.2 N/m, respectively.

Fig. 11 shows the comparisons of the simulated VIRs using the assumed mechanical damping and stiffness coefficients for the wind speed of 4.64 m/s (Case A in Figs. 11) and 5.14 m/s (Case B in Fig. 11) with the VIRs measured in the wind tunnel. Clearly, the CFD simulation can predict the variations of VIR with wind speed and lock-in regions, indicating that the numerical model developed by CFD simulation in this study can be a virtual model of the test deck section. Nevertheless, Fig. 11 also indicates that the VIR amplitude and lock-in region of the deck section predicted by the virtual model for Case A and Case B deviate from those measured in the wind tunnel. The measured global maximum displacement amplitude lies between Case A and Case B. The lock-in region of Case B becomes the narrowest. Clearly, the mechanical damping and stiffness coefficients significantly affect the VIR and lock-in region predicted by the virtual model. It is important to convert the virtual model to a digital twin by eliminating uncertainties through the fusion of wind tunnel test results with the virtual model to provide more accurate predictions for VIV.

5. Establishment of local digital twins within lock-in region

In this section, the method of developing a local digital twin for VIV at a specific wind speed using the multivariate polynomial regression algorithm is first presented. The method is then applied to establish a series of local digital twins at different wind speeds within the lock-in region.

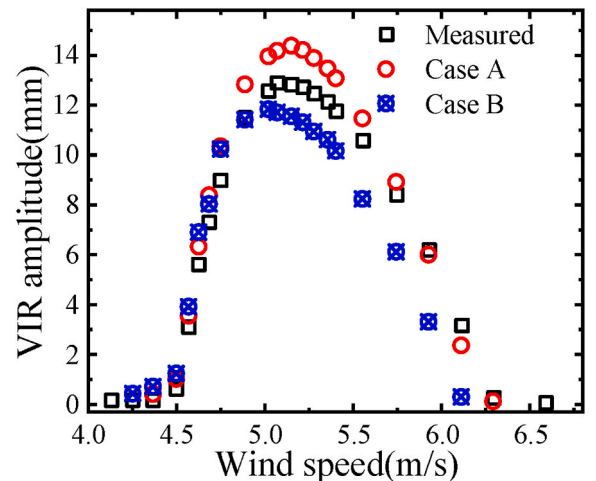


Fig. 11. Comparison of simulated VIRs with measured VIRs.

5.1. Strategies for establishing a local digital twin

The mechanical damping and stiffness coefficients of the deck section are selected as optimization parameters because of their significance on VIV and their uncertainties, as discussed before. The establishment of a local digital twin is thus aimed at finding out the most accurate mechanical damping and stiffness coefficients through the parameter updating to ensure the simulated VIR by the digital twin are as close as possible to those measured from the wind tunnel. In this regard, the minimization of relative errors in the VIR response, reflected by the vibration amplitude and frequency, between the measured and simulated ones to less than 1 % is taken as the threshold, which is named as the objective function. The relative error is expressed as:

$$R_r = \frac{|f_0^s - f_0^m|}{f_0^m} + \frac{|y^s - y^m|}{y^m} \quad (9)$$

where f_0^s and y^s are the simulated VIR frequency and VIR amplitude by the virtual model, while the f_0^m and y^m are the measured VIR frequency and VIR amplitude, respectively.

The optimization algorithm minimizes the relative error R_r by selecting appropriate parameter values. Since CFD simulations consume significant computational resources, the optimization algorithm is expected to satisfy the requirement of R_r less than 1 % with minimal iterations. Therefore, the multivariate polynomial regression of the highest quadratic term is adopted by considering its performance on optimization and the weak nonlinearities between the optimization parameters and VIR. The expression for this optimization algorithm is therefore given as:

$$c = a_1 + a_2 y^s + a_3 f_0^s + a_4 (y^s)^2 + a_5 (f_0^s)^2 + a_6 y^s f_0^s \quad (10a)$$

$$k = b_1 + b_2 y^s + b_3 f_0^s + b_4 (y^s)^2 + b_5 (f_0^s)^2 + b_6 y^s f_0^s \quad (10b)$$

where c and k are the mechanical damping and stiffness coefficients used in the virtual model; y^s and f_0^s are the VIR amplitude and VIR frequency predicted by the virtual model; a_i and b_i ($i = 1, 2, \dots, 6$) represent the coefficients of the multivariate polynomial regression for the mechanical damping and stiffness coefficients, respectively.

The establishment of a digital twin at a specific wind speed within the lock-in region is achieved through a joint simulation of MATLAB and FLUENT. The process is shown in Fig. 12 and described as follows: (1) determine the intervals for mechanical damping and stiffness coefficients to ensure the simulated maximum VIR amplitude and frequency exceed the values measured in the wind tunnel while the simulated minimum values remain lower than the measured values; (2) select initial samples of the mechanical damping and stiffness coefficients using the central composite design (CCD) method, which is capable of evaluating main, interaction, and curvature effects of optimization coefficients on VIV; (3) simulate VIV with the chosen samples in FLUENT; (4) construct the surrogate model (Eq. (10)) in MATLAB based on the simulation results; (5) use the surrogate model to predict the mechanical damping and stiffness coefficients of the deck section at the given wind speed; (6) conduct the CFD simulation in FLUENT with the predicted mechanical damping and stiffness coefficients; (7) extract the simulated VIR amplitude and frequency and calculate the relative error; (8) terminate the iteration if the relative error is smaller than the threshold ($R_r < 1\%$); otherwise, update the surrogate model with the VIR amplitude, frequency, and coefficients of the current iteration; and (9) repeat steps (5) to (8) until the calculated relative error meets the threshold of establishing a digital twin for VIV.

In Step (6), the simulation strategy of performing FSI from zero initial conditions to steady vibrations consumes a significant amount of computational time. Aiming at improving the efficiency of CFD simulation, the recorded VIR time history at the given wind speed obtained from the wind tunnel test is used to drive the numerical deck section

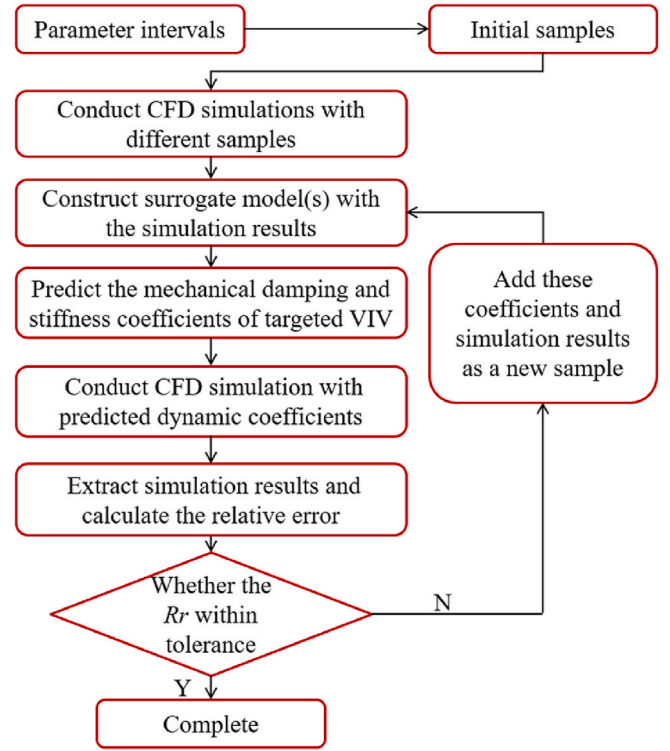


Fig. 12. The procedure of developing a local digital twin at a given wind speed.

until the flow fields surrounding the deck section become steady (Li et al., 2024a). Then, the weakly coupled CFD simulation takes over to conduct a fully FSI simulation. This computation strategy was used during the establishment of a digital twin.

5.2. Establishing a local digital twin at a given wind speed

By taking the VIV of the deck section at a wind speed of 5.14 m/s as an example, the establishment of a local digital twin at this wind speed is first introduced in this section. The local digital twins of the bridge deck at other test wind speeds are then established using the same way as used in the first local digital twin.

Based on the CFD simulation results presented in Fig. 11, the lower and upper limits for the mechanical damping coefficient are set as 42.55 N s/m and 53.16 N s/m, while the lower and upper limits for the mechanical stiffness coefficient are set as 146042.13 N/m to 152913.93 N/m, respectively. In this study, the Central Composite Design (CCD) is adopted to select the initial samples as it is the standard statistical design method for fitting second-order polynomials. By appropriately arranging central points, factorial points, and axial points, CCD enables unbiased estimation of quadratic and interaction terms and provides an effective assessment of curvature effects (Park et al., 2008). In the CCD method, the axial points are set at $\pm\alpha$ along each axis, where α is typically the square root of the number of experiments. In this two-level factorial design with two factors, α is selected as $\sqrt{2}$ (approximately 1.414), based on the fundamental test count of 2. The central mechanical damping coefficient is 47.85 N s/m and the central mechanical stiffness coefficient is 149478.03 N/m, which are the average values of their respective ranges. The axial points distances are 41.98 N s/m for the mechanical damping coefficient and 6871.8 N/m for the mechanical stiffness coefficient. The axial points positions, adjusted by ± 1.414 , include: the mechanical damping coefficients at 40.40 N s/m (lower) and 55.34 N s/m (upper); the mechanical stiffness coefficients at 144483.69 N/m (lower) and 154472.37 N/m (upper). As a result, the nine initial samples are detailed in Fig. 13 and Table 4.

The factorial design shown in Table 4 includes all combinations of

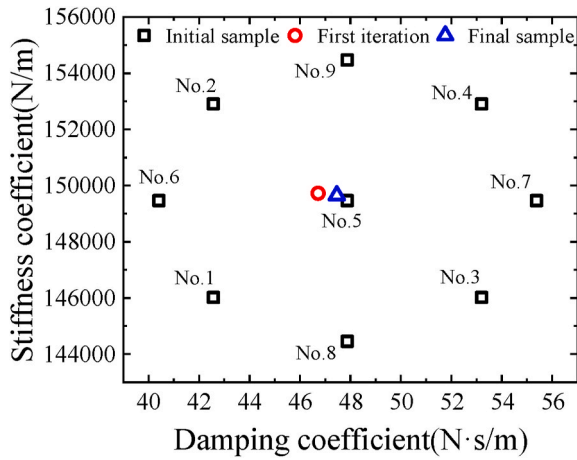


Fig. 13. The distribution of samples by CCD for wind speed of 5.14 m/s.

Table 4

Initial samples and its simulation results for establishing digital twin for wind speed of 5.14 m/s.

No.	Mechanical damping coefficient (N·s/m)	Mechanical stiffness coefficient (N/m)	Type	VIR amplitude (mm)	VIR frequency (Hz)
1	42.55	146042.13	Factorial point	14.06	4.283
2	42.55	152913.93	Factorial point	14.01	4.391
3	53.16	146042.13	Factorial point	11.61	4.267
4	53.16	152913.93	Factorial point	11.58	4.378
5	47.85	149478.03	Center point	12.84	4.331
6	40.40	149478.03	Axial point	14.54	4.338
7	55.34	149478.03	Axial point	11.26	4.324
8	47.85	144483.69	Axial point	12.79	4.261
9	47.85	154472.37	Axial point	12.77	4.411

the factors. This setup allows for the evaluation of the main effects, or linear effects, of each factor to observe the response changes caused by variations in a single factor. Additionally, combining different factor levels enables assessment of interaction effects, or how responses change when two factors vary simultaneously. As shown in Table 4, the axial points are positioned above or below the basic design levels along each factor axis. These points are essential for estimating the quadratic or curvature effects of each factor, representing nonlinear response influences. The inclusion of axial points enhances the design flexibility and enables the surrogate model to more accurately describe the overall trend of the response as the factor levels change.

Table 4 also presents the simulation results of VIR amplitudes and frequencies corresponding to the initial samples. It shows that the VIR amplitude varies from 11.26 mm to 14.54 mm, and the VIR frequency ranges from 4.261 Hz to 4.411 Hz. The simulated ranges of VIR amplitude and frequency cover the measured results of 12.89 mm and 4.333 Hz, indicating that the selected ranges of optimization parameters are appropriate. These samples and simulation results are then used to construct the surrogate model in MATLAB, following Eq. (10). The procedure of developing a local digital twin shown in Fig. 11 is then followed to iteratively update the virtual model until the relative error is smaller than 1 %. Table 5 shows that the digital twin established after

Table 5

Iteration results for wind speed of 5.14 m/s.

Iteration No.	Mechanical damping coefficient (N·s/m)	Mechanical stiffness coefficient (N/m)	VIR amplitude (mm)	VIR frequency (Hz)	R _r (%)
1	46.72	149725.29	13.06	4.337	1.41
2	47.46	149651.58	12.92	4.334	0.24

two iterations and the optimized mechanical damping and stiffness coefficients of 47.46 N s/m and 149651.58 N/m, respectively, for the bridge deck under a wind speed of 5.14 m/s. Moreover, it was also illustrated in Fig. 13 for comparison with the initial samples.

Fig. 14(a and b) show the VIR and aerodynamic lift force time histories of the digital twin at a wind speed of 5.14 m/s. During the first 20 cycles, the mapping vibration method dominates the simulation, and then the FSI takes over.

Fig. 15(a and b) show the comparisons of the VIR time history and its power spectral density (PSD) of the digital twin with those of the physical model at the wind speed of 5.14 m/s. The good agreements demonstrate the capability of the digital twin in replicating the VIV of the physical model. Fig. 15(c and d) show the corresponding comparisons of the aerodynamic lift force time history and its PSD of the digital twin. The comparisons further verify the accuracy of the digital twin, as the measured aerodynamic lift force was not used during the optimization process.

In the same way, the local digital twins of the deck section at other test wind speeds can be established. The optimized results and the final sampling points together with the corresponding relative errors are presented in Table 6 and Fig. 16. As shown in Table 6, the developed local digital twins accurately reproduce VIV. Table 6 also highlights that the mechanical damping coefficients of the deck section vary significantly from 30.11 N s/m at the wind speed of 4.69 m/s to 45.98 N s/m at the wind speed of 5.14 m/s, an increase of 52.7 %. However, the mechanical stiffness coefficients of the bridge deck exhibit a small variation.

The fitted relationships of the mechanical damping and stiffness coefficients with the VIR amplitudes are shown in Fig. 17. The fitted equations are as follows:

$$c = 17.605 + 2.20y \quad (11a)$$

$$k = 153202.18 - 274.87y \quad (11b)$$

where the c and k are the mechanical damping coefficient and the stiffness coefficient of the bridge deck, respectively; and y is the vibration amplitude.

6. Predictions by local digital twins

The local digital twins established in Section 4 for VIV within the lock-in region are used to explore the blockage effects. These local digital twins are then utilized to simulate the VIV of the deck section of other mechanical damping levels, as measured in the wind tunnel and shown in Fig. 6.

6.1. Blockage effects on VIV

As discussed before, the VIV of the deck section in a wind tunnel is influenced by air-induced damping and mass, and such an effect is related to the relative size of the wind tunnel to the deck section. The blockage ratio, defined as the proportion of the projected deck section area to the cross-sectional area of the wind tunnel, is used here to assess this effect. The blockage ratio for this wind tunnel test is calculated to be 3.88 % without incoming wind and 4.46 % at a wind speed of 5.14 m/s.

To investigate the blockage effect, the computational domain of the

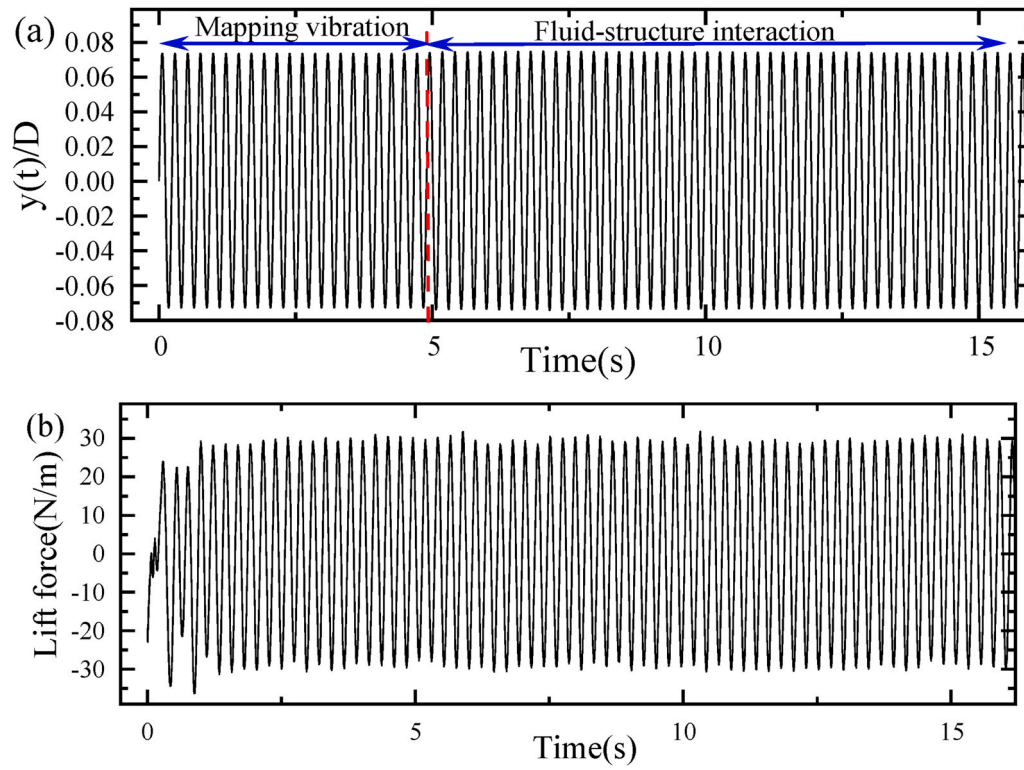


Fig. 14. Time histories of the digital twin: (a) VIR, (b) aerodynamic lift force.

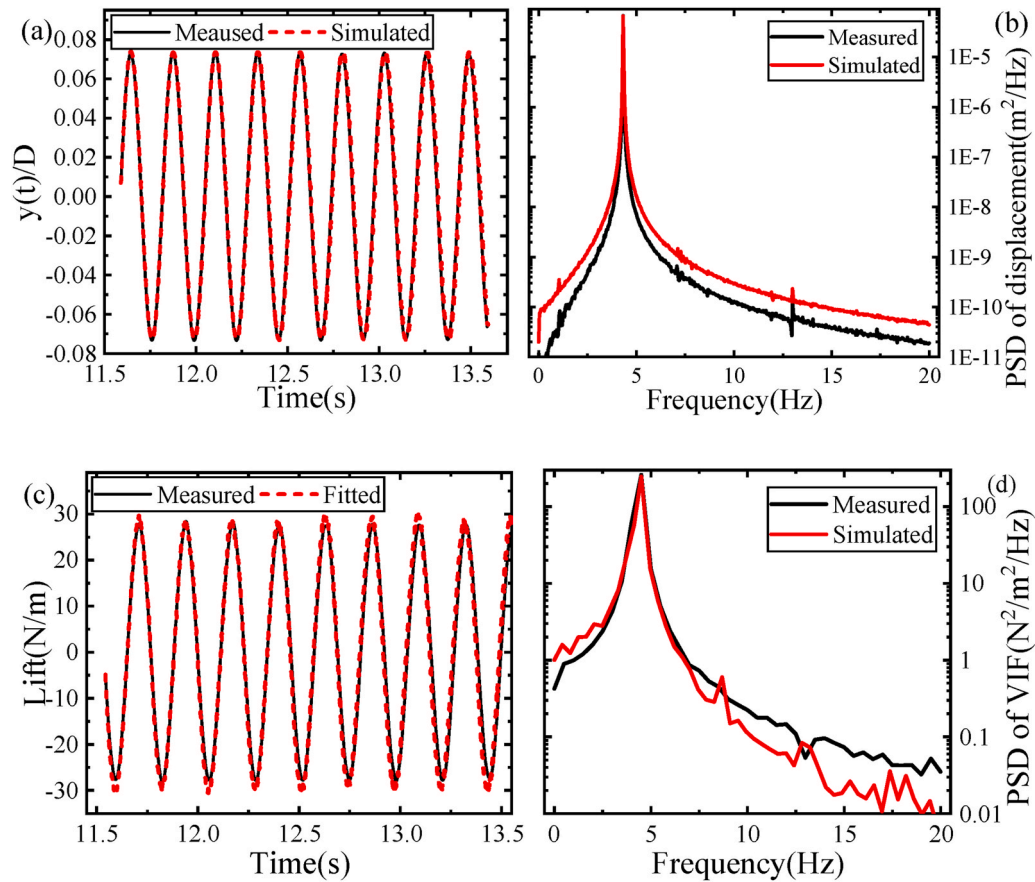


Fig. 15. Comparisons of VIR and VIF between the digital twin and physical model at 5.14 m/s: (a) VIR time histories, (b) PSD of VIR, (c) aerodynamic lift force time histories, (d) PSD of aerodynamic lift force.

Table 6

Local digital twins established for VIV of the bridge deck under different wind speeds.

No.	Wind speed (m/s)	VIR amplitude (mm)	VIR frequency (Hz)	Mechanical damping coefficient (N·s/m)	Mechanical stiffness coefficient (N/m)	R_r (%)
2	4.69	5.63	4.323	30.108	151709.61	0.36
4	4.82	9.04	4.328	37.44	150697.56	0.58
5	4.95	11.47	4.331	42.861	150107.49	0.33
7	5.14	12.92	4.333	45.981	149651.58	0.23
9	5.28	12.68	4.335	45.396	149727.24	0.24
11	5.42	12.19	4.342	44.343	149835.66	0.46
13	5.62	10.55	4.353	40.833	150257.64	0.57
15	5.99	6.23	4.377	31.356	151445.97	0.51

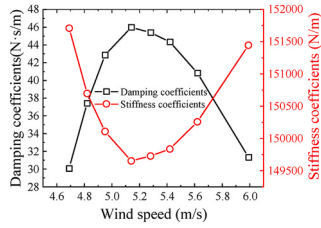


Fig. 16. The final samples used for constructing the local digital twin within the lock-in region.

digital twin, as illustrated in Fig. 9, is expanded by a factor of 5 from the center of mass of the bridge section (see Fig. 18): the computational domain is extended from 12.3B to 61.5B in the length direction and from 4.5m to 22.5m in the height direction, resulting in a blockage ratio of 0.78 %. Nevertheless, while the dimensions of Zone 1 (stationary zone) are increased, the dimensions of Zone 2 (deformative zone) and Zone 3 (rigid zone) remain unchanged. The change of meshes in Zone 1 of the computational domain is implemented following the original criteria and using ANSYS ICEM software. The boundary conditions for the upper

and bottom walls are also changed to symmetry to further reduce the blockage effect.

To ensure that the VIV of the bridge deck simulated from the digital twin is independent of the size of the computational domain, the blockage ratios of 1 % and 0.39 % are also considered. For the wind speed of 5.14 m/s, the mechanical damping coefficient of 41.23 N s/m, and the stiffness coefficient of 147831.80 N/m, the stable peak VIR amplitudes of the deck section in the three expanded computational domains are found to be 14.24 mm with the blockage ratio of 1 %, 14.81 mm with 0.78 %, and 14.82 mm with 0.39 %, respectively. These results indicate that the computational domain at the blockage ratio of 0.78 % is large enough to mitigate the blockage effect.

The CFD simulations are then performed using this digital twin with the optimized mechanical damping coefficients and stiffness coefficients at the associated wind speeds. The simulation results, along with the measurement results with the blockage ratio of 3.88 %, are illustrated in Fig. 19. As can be seen, although the variation of VIR amplitude with wind speed is similar between those predicted by the digital twin with the blockage ratio of 0.78 % and those measured in the wind tunnel with the blockage ratio of 3.88 %, the maximum predicted VIR amplitude is still 3.34 % larger than that from the wind tunnel. This indicates that the blockage ratio of 3.88 % still underestimates the maximum vibration amplitude, even though it is less than 5 %. Therefore, the digital twin shall be used to eliminate the blockage effect if the blockage ratio in the wind tunnel test is larger than 5 %.

6.2. Predicting VIV of deck section of different mechanical damping levels

The VIV tests of the twin-box bridge deck were conducted for not only NAD case but also other three damping levels. However, the variations of mechanical damping and stiffness coefficients with VIR amplitude of the deck section of other three damping levels are unknown. The method presented in Section 4 can be employed to establish the local digital twins of VIV for the deck section with other three damping levels but it requires large computational resources. This subsection uses the local digital twins established for the deck section

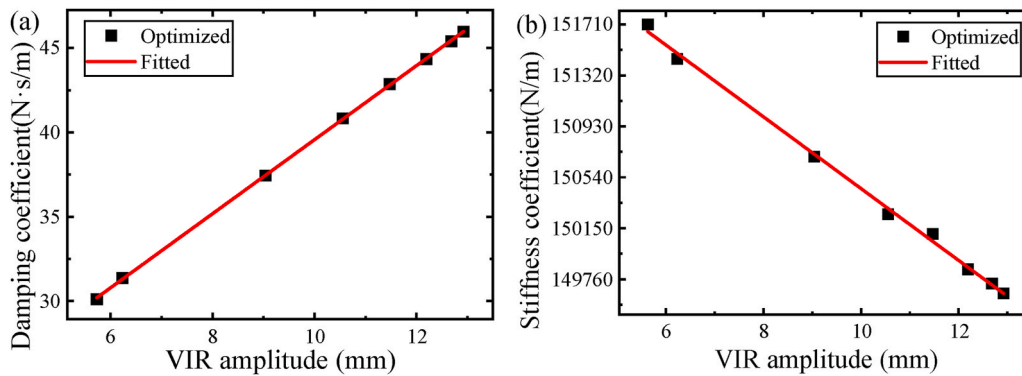


Fig. 17. The fitted relationships of mechanical damping and stiffness coefficients with vibration amplitude of deck section of NAD.

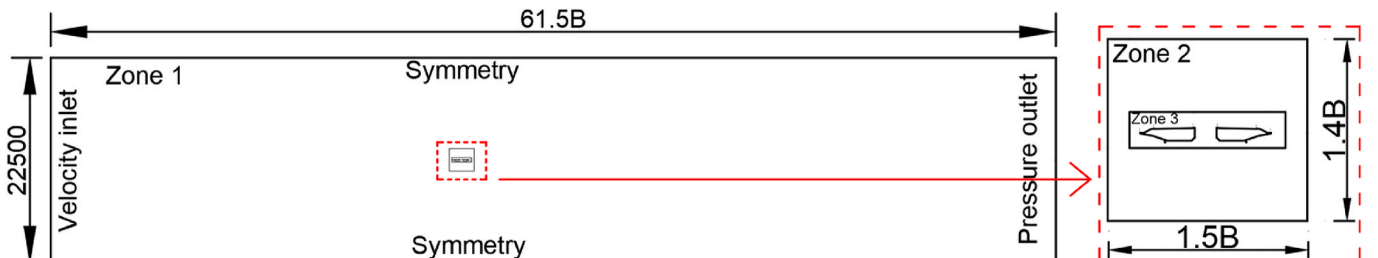


Fig. 18. The computational domain to eliminate the blockage effect (unit: mm).

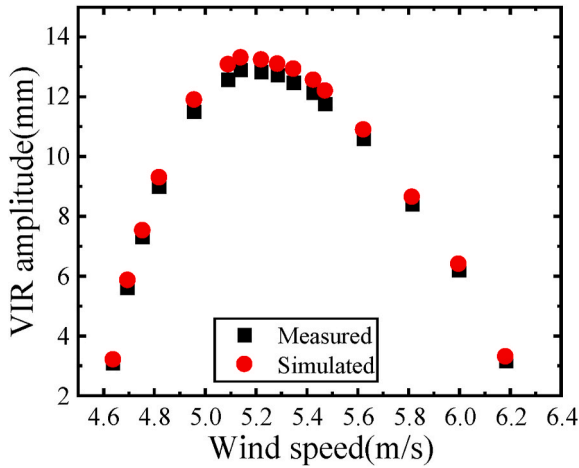


Fig. 19. VIRs predicted by digital twin at a blockage ratio of 0.78 % and measured from physical model at a blockage ratio of 3.88 %.

with NAD to indirectly obtain the local digital twins of the deck section with other three damping levels.

The variations of the mechanical stiffness coefficients with vibration amplitude of the deck section with NAD have been obtained (see Eq. (11b)). Therefore, with Eq. (4b), Eq. (6a), and Eq. (11b), the relationship between the air-induced mass (m_0) and vibration amplitude can be obtained:

$$m_0 = \frac{153202.18 - 274.87y}{4\pi^2(4.314 - 0.00335y + 0.0000794y^2)^2} - m \quad (12)$$

where y represents the vibration amplitude; and m is the mass of the deck section.

Since the air-induced mass of the deck section depends only on its aerodynamic shape and size of the deck section in the given wind environment (Zhang and Xu, 2019), the relationship between m_0 and vibration amplitude expressed by Eq. (12) is also applicable to the deck section with the other damping level. As a result, the variation of the mechanical stiffness coefficient with vibration amplitude of the deck section of other mechanical damping levels can be obtained by Eq. (4b). Furthermore, as the air-induced mass and the variation of the mechanical damping coefficient with the vibration amplitude of the deck section of NAD are known, the air-induced damping coefficient of the spring-suspended twin-box deck of NAD is obtained using Eq. (13):

$$c_{air}(y) = 4\pi f_{0,NAD}(y)\xi_{0,NAD}(y)(m + m_0) - 2.20y - 17.605 \quad (13)$$

In the given wind environment, the air-induced damping coefficient of the deck section also depends only on its aerodynamic shape and size. Thus, the identified air-induced damping coefficient from Eq. (13) is also applicable to the deck section of other mechanical damping levels. By converting the nominal damping ratio of AD(i) in Eq. (5) to the nominal damping coefficient through Eq. (4a) and subtracting the air-induced damping coefficient, the variation of the mechanical damping coefficient with the vibration amplitude of the deck section of other damping levels can be obtained by:

$$c_{ADi} = 4\pi f_{0,ADi}(y)\xi_{0,ADi}(y)(m + m_0) - c_{air}(y) \quad (14)$$

where the subscript AD(i) represents the additional damping level i .

By using the measured VIR amplitudes of the deck section with other three damping levels, the mechanical damping and stiffness coefficients of the deck section with other three damping levels can be obtained from Eq. (14) and Eq. (4b). Then, the CFD simulations of the deck section with other three damping levels are performed by using the mechanical damping and stiffness coefficients obtained and the corresponding wind speeds to obtain the VIR amplitudes and the associated local digital

twins.

The VIR amplitudes of the deck section with other three damping levels predicted by the local digital twins are compared with those measured in the wind tunnel, as shown in Fig. 20. It is seen that the local digital twins accurately predict the lock-in region and the variations of VIR amplitude with wind speed for the deck section with different damping levels. The relative errors between the predicted VIR amplitudes and the measured VIR amplitudes range from 1 % to 2 % when the VIR amplitude of the deck section is larger than 4 mm. When the VIR amplitudes are smaller than 4 mm, the relative errors are slightly large due to the small VIR amplitudes but remain less than 6 %.

7. Global digital twins

In this section, global digital twins are established based on the local digital twins and Kriging interpolation. These global digital twins are used to predict VIV within the lock-in region.

7.1. Establishment of a global digital twin for deck section with NAD

This subsection constructs a global digital twin (a surrogate model) using the Kriging interpolation and the local digital twins. This global digital twin represents the general relationship between the wind speed (input) and VIR amplitude (output) of the deck section within the lock-in region.

The Kriging interpolation (Krige, 1951; Sacks et al., 1989) is a powerful tool for spatial statistics and interpolation. It not only allows for interpolation between known points but also provides uncertainty estimates for predictions. The corresponding equation used to predict the VIR amplitude for any wind speed within the lock-in region is given by:

$$\hat{y}(u) = f(u) + Z(u) \quad (15)$$

where $\hat{y}(u)$ is the predicted VIV amplitude at wind speed u within the lock-in region. The trend function $f(u)$ reflects the overall variation of amplitude with wind speed, while the Gaussian term $Z(u)$ acts as a zero-mean process that captures the correlation among sample points, thereby compensating for the part that cannot be captured by the trend function.

The selection of the trend function is typically determined from the relationship between the inputs and outputs in the training samples, and it may take constant, linear, or polynomial forms. In this study, based on the relationship between the wind speed and VIV amplitude shown in Table 1, the following expression is adopted as the trend function:

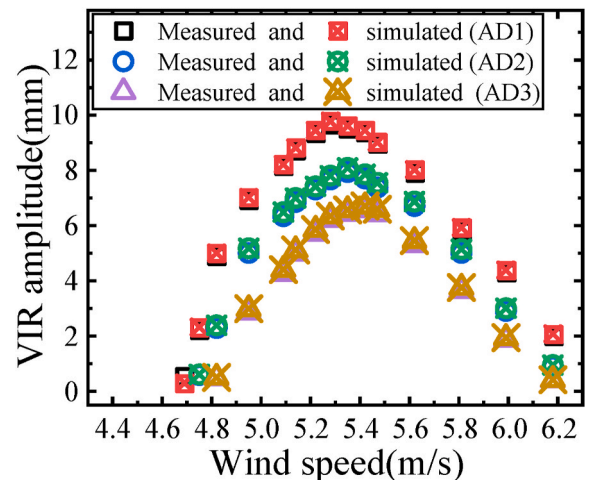


Fig. 20. Prediction of VIRs for different mechanical damping levels with the digital twin.

$$f(u) = \beta_0 + \beta_1 u + \beta_2 u^2 + \beta_3 u^3 + \beta_4 u^4 \quad (16)$$

where $\beta_i (i = 1, 2, 3, 4)$ are the parameters of the trend function, which are estimated from the sample data.

To accommodate the local fluctuations and nonlinear responses in the wind speed–vibration amplitude relationship, a zero-mean Gaussian term is incorporated to model the residuals and thereby allow smooth interpolation among the available samples. The Gaussian term is characterized by the correlation structure among the training samples. In this study, the following correlation function is adopted:

$$R(u_i, u_j) = \exp(-\theta(u_i - u_j)^2) \quad (17a)$$

where u_i and u_j are the wind speeds of the i -th and j -th training samples, and θ is the scale parameter controlling the rate at which the correlation decays with increasing wind speed difference. On this basis, the $n \times n$ correlation matrix $R = [R(u_i, u_j)]_{i,j=1}^n$ can be assembled. By scaling this matrix with the process variance σ^2 , the covariance matrix is obtained as:

$$K = \sigma^2 R \quad (17b)$$

which encapsulates the joint dependence among the training samples and forms the basis of Gaussian process prediction.

Accordingly, the Gaussian term at the prediction point takes the form:

$$z(u) = r(u)^T R^{-1} (y - f(u)) \quad (18)$$

where $r(u)$ is the correlation vector between the prediction point and the training points, R^{-1} is the inverse of the correlation matrix, $y = [y(u_1), y(u_2), \dots, y(u_n)]^T$ is the vector of vibration amplitudes at the training samples, and $f(u)$ denotes the trend function.

The hyperparameters of the global digital twin (β , σ^2 , θ) are estimated from the training samples using the maximum likelihood method. Given its extensive application in bridge wind engineering and the maturity of its theoretical foundations, the detailed derivation of the hyperparameter estimation is omitted here for brevity, with references provided for further details (Kriging, 1951; Sacks et al., 1989).

The global digital twin is constructed using the 16 valid wind speeds and corresponding VIV responses listed in Table 1, with 8 samples (Table 6) for training and the remaining 8 for validation. During training, the trend function is specified according to Eq. (16), and the parameters of the trend function together with the hyperparameters of the Gaussian process are estimated by the maximum likelihood method. On this basis, the predictive model is established and the global digital twin is constructed. The validation wind speeds are then input into the

model to obtain predicted amplitudes, which are compared with the test results to evaluate its prediction ability.

Fig. 21a shows the VIR amplitudes of the deck section with NAD measured from the wind tunnel, simulated by the local digital twins, and predicted by the global digital twin within the lock-in region. It is seen that both the lock-in region and the VIR amplitudes of the deck section predicted by the global digital twin match well with the measured ones. Fig. 21b shows the relative errors between the VIR amplitudes predicted by the global digital twin and those measured from the wind tunnel. The relative errors between the VIR amplitudes simulated by the local digital twins and those measured from the wind tunnel are also shown in Fig. 21b. It is seen that the global digital twin provides satisfactory prediction of the VIR amplitudes, with all relative errors smaller than 1 %, except for the VIR amplitudes at the first and last wind speeds. The predictions at these two wind speeds involve the extrapolation of the Kriging interpolation, which could lead to relatively larger errors. Moreover, it is also because the VIR amplitudes at these two wind speeds are so small that even a slight difference in absolute amplitude can result in a relatively large error. Considering these factors, it can be concluded that the established global digital twin can provide a reliable and accurate prediction of the VIR amplitudes of the deck section within the entire lock-in range.

7.2. Establishment of a global digital twin for deck section with different damping levels

The global digital twin developed in Section 6.1 is for the prediction of VIR amplitude of the deck section with NAD only. The global digital twin to be established in this subsection is for the prediction of VIR amplitude of the deck section with different damping systems.

The inputs to the Kriging interpolation this time are wind speed, mechanical damping and stiffness coefficients corresponding to the four damping levels (58 samples), while the output is VIR amplitude. Since using a fourth-order polynomial as the trend function would introduce many high-order interaction terms and risk overfitting with limited samples. To avoid this, a quadratic polynomial with squared and interaction terms is adopted:

$$f(u, c, k) = \beta_0 + \beta_u u + \beta_c c + \beta_k k + \beta_{uu} u^2 + \beta_{cc} c^2 + \beta_{kk} k^2 + \beta_{uc} uc + \beta_{uk} uk + \beta_{ck} ck \quad (19)$$

where $\beta = (\beta_0, \beta_u, \beta_c, \beta_k, \beta_{uu}, \beta_{cc}, \beta_{kk}, \beta_{uc}, \beta_{uk}, \beta_{ck})^T$ denotes the parameters of the trend function to be estimated. The quadratic terms in the trend function capture the nonlinear curvature, while the interaction terms reflect the coupling effects among the parameters. Based on this

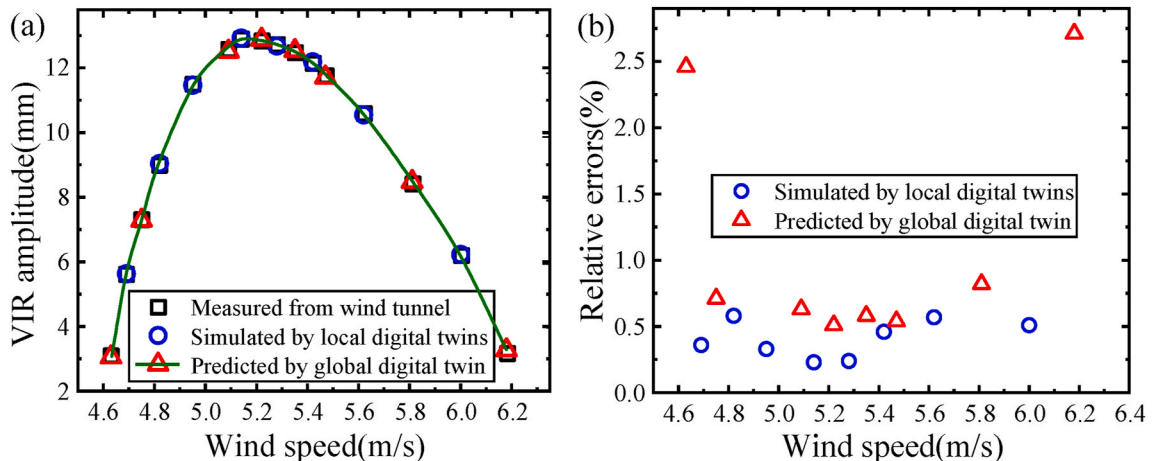


Fig. 21. Prediction of VIR amplitudes of deck section with NAD by global digital twin.

trend function, the procedure for constructing the global digital twin remains consistent with that described in Section 7.1.

To demonstrate the prediction capacity of this global digital twin, a new set of mechanical damping coefficients are assumed by averaging the mechanical damping coefficients of the deck section for NAD and AD3 damping levels, and the VIR amplitudes of the deck section with the new set of mechanical damping coefficients are predicted by this global digital twin. The predicted results are shown on Fig. 22. It is seen that the predicted VIR amplitude falls between the VIR amplitudes of the deck section with NAD and AD3 at each wind speed, which confirms the reliability of this global digital twin. Moreover, by using the leave-one-out cross-validation method, the reliability of this global digital twin is further validated and the relative errors are about 2.7 % and the maximum relative error is smaller than 4.6 %.

8. Conclusions

Local digital twins and global digital twins have been established in this study for VIV prediction of a twin-box bridge deck section within its lock-in region by mapping VIV test results from the wind tunnel with CFD simulations. The specific conclusions from this study are summarized as follows:

- (1) The damping ratio and vibration frequency of the bridge deck section identified from the free-decay curve cause uncertainties in the mechanical damping and stiffness of the bridge deck due to their nonlinearities as well as air-induced damping and mass, which further leads to the discrepancies between CFD-simulated and wind tunnel tested VIV responses.
- (2) Local digital twins for VIV of the bridge deck section within the lock-in region are established by mapping wind tunnel results to CFD simulations through multivariate polynomial regression. By iteratively updating, they eliminate uncertainties in mechanical damping and stiffness, ensuring that the simulated VIV amplitudes agree well with the wind tunnel tests.
- (3) The predictions made by the local digital twins indicate that the wind tunnel tests may underestimate the VIR amplitudes due to blockage effect. The local digital twins can be used to effectively eliminate the blockage effects on VIV.
- (4) The air-induced damping and air-induced mass of the deck section in the wind tunnel are identified and the VIV amplitude of the deck section at different damping levels are predicted by the local digital twins.
- (5) A global digital twin is developed by using the local digital twins of the deck section under NAD damping system and the Kriging interpolation. This global digital twin could provide an accurate prediction of VIV amplitude of the deck section at any wind speed within the lock-in region.
- (6) Another global digital twin is also developed by using the local digital twins of the deck section with different damping levels and the Kriging interpolation. The established global digital twin could provide accurate predictions of VIR amplitudes of the deck section at any wind speed, mechanical damping coefficient, and mechanical stiffness coefficient of the deck section within the lock-in region.

The global digital twin developed in this study achieves satisfactory accuracy in predicting VIV. Future work will consider expanding the sample size of damping levels to further improve prediction robustness and extend the model's applicability to more refined tasks.

It should be noted that the digital twins developed in this study are based on vertical VIV of a twin-box girder model and therefore they cannot be directly applied to torsional VIV or other bridge deck sections subjected to VIV. Nevertheless, the proposed framework is generalizable and can be used to develop the digital twin for torsional VIV or other deck sections subjected to VIV.

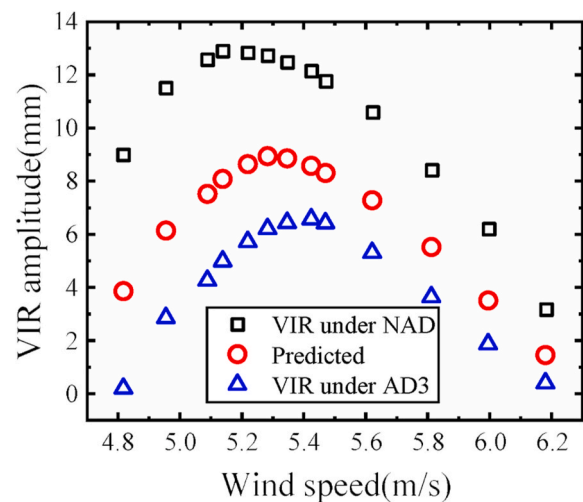


Fig. 22. The predicted VIR amplitudes by the global digital twin for different damping levels.

Furthermore, although this study addresses several aerodynamic issues in the VIV of the bridge deck, other aspects such as turbulence effects (Zhao et al., 2023) and VIV mechanisms (Hu et al., 2022) lie beyond its scope and are not discussed. These issues may be explored in future studies using the proposed digital twin framework.

CRediT authorship contribution statement

Hao-Yang Li: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **You-Lin Xu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Bo-Man Cheng:** Software, Methodology, Investigation. **Shang-Jun Jiang:** Writing – original draft, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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